Hadi Mohammadi

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Profile

Researcher in Explainable Natural Language Processing for social-science applications, having completed my PhD in the Department of Methodology & Statistics at Utrecht University. Across six doctoral research projects, I develop transparent and reliable methods for large language models, covering LLM-as-judge evaluation, annotation quality, cultural alignment, and token-level robustness, and connect them to applied data science questions. I work as an independent researcher who leads projects end to end: alongside the PhD I have led five competitively funded projects, taking each from proposal through to delivery. I now serve as Senior AI & Data Science Expert at AcademicTransfer, where I lead applied LLM work in production and am technical lead of three applied AI projects. I work comfortably across statistics, machine learning, and applied data science, and am committed to open, reproducible research, releasing code and data with my work.

Education

Utrecht University — Utrecht, The Netherlands
Ph.D. *in Explainable Natural Language Processing for Social Sciences* — *Feb 2023 – May 2026 (three years)*
Faculty of Social and Behavioural Sciences, Department of Methodology & Statistics

Thesis: *Let Me Explain! Explainable NLP for Understanding Large Language Models*

Member of: M&S Natural Language and Text Processing (*NLTP*) group and Human Data Science (*HDS*) Lab

Supervisors: Dr. Robert A. Bagheri, Dr. Anastasia Giachanou, and Prof. Dr. Daniel Oberski

Projects:

- Explainability in Practice: A Survey of Explainable NLP Across Domains
- A Transparent Pipeline for Identifying Sexism in Social Media
- Explainability-Based Token Replacement on LLM-Generated Text
- Assessing the Reliability of LLM Annotations under Demographic Bias and Model Explanation
- Exploring Cultural Variations in Moral Judgments with Large Language Models
- EvalMORAAL: Interpretable Chain-of-Thought and LLM-as-Judge Evaluation for Moral Alignment

University of Tehran — Tehran, Iran
M.Sc. *in Industrial Engineering, Macro Systems;* ***GPA: 18.96/20 (4/4)*** — *Sep 2018 – Apr 2021*

Thesis: "Dynamic pricing with demand learning using reinforcement learning in a non-stationary environment by considering customer behaviour and advertisement (a data-driven approach)."

Cognitive Science Researcher: Conducted research in collaboration with the *Institute for Cognitive Science Studies (ICSS)* on customer behaviour in dynamic pricing, analysing how strategic customers postpone purchases by comparing expected future value with that of myopic customers.

Isfahan University of Technology, Golpayegan College of Engineering — Isfahan, Iran
B.Sc. *in Mechanical Engineering* — *Sep 2011 – Jul 2016*

Publications

Mohammadi, H., Giachanou, A., & Bagheri, R. A. (2026). EvalMORAAL: Interpretable Chain-of-Thought and LLM-as-Judge Evaluation for Moral Alignment in Large Language Models. *Proceedings of the 15th Joint Conference on Lexical and Computational Semantics (*SEM 2026), ACL 2026* (in press).

Mohammadi, H., & Bagheri, R. A. (2026). Exploring Cultural Variations in Moral Judgments with Large Language Models. *Computational Linguistics in the Netherlands Journal* (in press).

Mohammadi, H., Kozak, T., & Giachanou, A. (2025). Evaluating GRPO and DPO for Faithful Chain-of-Thought Reasoning in LLMs. *arXiv preprint.* arXiv:2512.22631.

Mohammadi, H.†, Shahedi, T.†, Mosteiro, P., Poesio, M., Bagheri, R. A., & Giachanou, A. (2025). Assessing the Reliability of LLM Annotations in the Context of Demographic Bias and Model Explanation. *Workshop on Gender Bias in Natural Language Processing (GeBNLP), ACL 2025*, pp. 92–104. doi:10.18653/v1/2025.gebnlp-1.9. †Equal contribution.

Mohammadi, H., Meijer, Y. F. S. S., Papadopoulou, E., & Bagheri, R. A. (2025). Do Large Language Models Understand Morality Across Cultures? *Proceedings of the 2nd LUHME Workshop*, pp. 30–39.

Mohammadi, H., Giachanou, A., Oberski, D. L., & Bagheri, R. A. (2025). Explainability-Based Token Replacement on LLM-Generated Text. *arXiv preprint.* arXiv:2506.04050.

Mohammadi, H., Bagheri, R. A., Giachanou, A., & Oberski, D. L. (2025). Explainability in Practice: A Survey of Explainable NLP Across Various Domains. *arXiv preprint.* arXiv:2502.00837.

Oberski, D. L., Domenech, J., Garcia-Bernardo, J., Gerbrands, P., Hassink, W., **Mohammadi, H.**, Mustica, P., Schilpzand, R., & Witteloostuijn, A. (2025). Measuring Firm Characteristics with Natural Language Processing (NLP) Models: An error-mitigating framework with tutorial guidance. *Strategic Organization* (under revision).

Mohammadi, H., Rahmati, M., & Shahedi, T. (2024). Novel Approaches in Financial Fraud Detection: Hybrid Machine Learning and Uncertainty-Based Deep Learning. *Proceedings of the 36th Benelux Conference on Artificial Intelligence and the 33rd Benelux Conference on Machine Learning (BNAIC/BeNeLearn 2024)*, Utrecht, The Netherlands.

Mohammadi, H., Giachanou, A., & Bagheri, R. A. (2024). A Transparent Pipeline for Identifying Sexism in Social Media: Combining Explainability with Model Prediction. *Applied Sciences, 14*(19), 8620. doi:10.3390/app14198620.

Fivez, P., Daelemans, W., Van de Cruys, T., Kashnitsky, Y., Chamezopoulos, S., **Mohammadi, H.**, Giachanou, A., Bagheri, R. A., Poelman, W., Vladika, J., Ploeger, E., Bjerva, J., Matthes, F., & van Halteren, H. (2024). The CLIN33 Shared Task on the Detection of Text Generated by Large Language Models. *Computational Linguistics in the Netherlands Journal, 13*, 233–259.

Mohammadi, H., Giachanou, A., & Bagheri, R. A. (2023). Towards Robust Online Sexism Detection: A Multi-Model Approach with BERT, XLM-RoBERTa, and DistilBERT for EXIST 2023 Tasks. In *Working Notes of CLEF 2023*, CEUR-WS Vol-3497, pp. 1000–1011.

Mohammadi, H. (2022). *Statistical Reinforcement Learning with Application in Dynamic Pricing Problem: Learning from Customer Interaction based on Bayesian Approach*. Tehran: Arshadan Publication. [In Persian]

Mirabnejad, M., **Mohammadi, H.**, Mirzabaghi, M., Aghsami, A., Jolai, F., & Yazdani, M. (2022). Home Health Care Problem with Synchronization Visits and Considering Samples Transferring Time: A Case Study in Tehran, Iran. *International Journal of Environmental Research and Public Health, 19*(22), 15036. doi:10.3390/ijerph192215036.

Grants & Awards

Best Oral Presentation: IOPS Winter Conference 2025 on *Cultural Moral Alignment in Large Language Models* (December 2025).

The focus area Applied Data Science Grant (Utrecht University), €5K: Project "Optimizing the CV Priority Sorter for Fair and Efficient Prioritization," in collaboration with Dr. Robert A. Bagheri and Dr. Georg Krempl (Utrecht University) and Jeroen Sparla (AcademicTransfer); December 2025.

The focus area Applied Data Science Grant (Utrecht University), €5K: Project "Rule Extraction from Human Evaluation Using LLMs and Reinforcement Learning," in collaboration with Dr. Anastasia Giachanou and Dr. Shihan Wang (July 2025).

Centre for Unusual Collaborations (CUCo) Spark Grant, €9K: Project "ReDOSE: Towards a More Circular Pharmaceutical Industry Using AI" (May 2025), in collaboration with Dr. Negin Salimi (WUR), Dr. Shayegheh Ashourizadeh (WUR), Dr. Duru Bayram (TU/e), and Dr. Saeed Arbabi (UMCU).

European Lighthouse to Manifest Trustworthy and Green AI (Enfield), €14.4K: Project "User Perspectives on Explainable AI" (December 2024).

The focus area Applied Data Science Grant (Utrecht University), €7.5K: Project "Assessing Reliability of Annotations in the Context of Model Predictions and Explanations," in collaboration with Dr. Pablo Mosteiro Romero, Dr. Anastasia Giachanou, and Prof. dr. Massimo Poesio (December 2023).

Scholarship for Admission: Explainable AI (XAI) Summer School, TU Delft, The Netherlands (Aug 2022).

Ranked 2nd: Among more than 50 Industrial Engineering graduate students at the University of Tehran (Apr 2021).

Best Paper Award: Second National Conference of the Iranian System Dynamics Society (2019).

Ranked 35th: National Iranian university entrance exam for Master of Industrial Engineering, among more than 10,000 participants (Sep 2018).

Leadership & Representation

Co-founder of Young Scholar Network for NLP@U: Established this network in November 2024 to unite PhD candidates, postdocs, and junior researchers working on NLP and text mining across Utrecht University and UMC Utrecht.

LERU Doctoral Summer School Representative: Selected as the representative of the Faculty of Social and Behavioural Sciences, Utrecht University, for the 2025 LERU Doctoral Summer School.

Teaching Experience

Transformers Workshop, University Medical Center Utrecht — Utrecht, The Netherlands
Practical Assistant — *Dec 2023*

- Supervised practical sessions in Python alongside Dr. Robert A. Bagheri and Prof. Dr. Albert Gatt.

Data Wrangling and Data Analysis, Utrecht University Utrecht, The Netherlands
Computer Lab Assistant *Sep 2023 – Oct 2023*

- Assisted in the Master in Applied Data Science (*ADS*) course, focusing on R for data visualisation, exploratory data analysis, supervised learning, and text mining.

Deep Learning Summer School, Neuromatch Academy Los Angeles, USA (Remote)
Lead Teacher Assistant *Aug 2021 – Sep 2021*

- Facilitated learning for an international student cohort and coordinated with mentors and organisers.

Supervision

Mijntje Meijer, Evi Papadopoulou, Lourenço Santos Moitinho de Almeida (ADS master theses, co-supervised with Dr. Robert A. Bagheri, 2024). Hein Brouwer, Tamás Kozák, Mohamad Jouhar (ADS master theses, co-supervised with Dr. Anastasia Giachanou, 2025). Aliyah Vos, Geanina Verestiuc, Thomas Rietman (ADS master theses, co-supervised with Dr. Robert A. Bagheri, 2026).

Selected Work Experience

AcademicTransfer (Academic job-boarding company) Utrecht, The Netherlands – Full-time
Senior AI & Data Science Expert *Dec 2025 – Present*

- Lead AI strategy and applied data-science delivery for academic hiring, translating research on explainable and reliable NLP into production features.
- Technical lead of a recognition-and-rewards policy analysis examining how promotion and reward policies are reflected in academic positions, with Dr. Sanli Faez (national manager, Recognition & Rewards, Universiteiten van Nederland) and university policy makers across the Netherlands.
- Technical lead of a new AI intelligence layer and applicant tracking system for AcademicTransfer, used by 22 Dutch research universities and university medical centres: advanced AI spanning CV creation and revision, semantic search, talent-pool matching, and best-candidate identification.
- Developing a new e-learning programme with the University of Amsterdam for PhD candidates and researchers.

AcademicTransfer (Academic job-boarding company) Utrecht, The Netherlands – Part-time
AI & Data Science Expert *Jan 2024 – Dec 2025*

- Developed *CV Priority Sorter* using Active Learning and Large Language Models (LLMs), cutting HR hiring time by 20%, in collaboration with Utrecht University, TU Delft, and Maastricht University.
- Built end-to-end generative-AI pipelines for automatic insight extraction and de-identification from CVs, improving candidate–job alignment accuracy by 17% and halving manual effort.
- Deployed transformer-based models on Azure ML with MLflow, Docker, and CI/CD; used A/B tests, uplift analysis, and bandit simulations to evaluate and improve ranking and recommendation models.

Utrecht University, School of Economics Utrecht, The Netherlands
Researcher & NLP Developer in the FIRMBACKBONE project *Feb 2024 – Mar 2024*

- Developed *fbb-sustainability-analysis-cli*, adding a keyword extractor for sustainability terms and an index generator that scores websites on sustainability content.

SnowaTec (Innovation Center) – Entekhab Electronic Tehran, Iran – Full-time
Senior Data Scientist in Customer Behaviour *Dec 2021 – Dec 2022*

- Managed the enterprise data warehouse with modern ETL tooling, improving data accessibility by 16%.
- Led the project to define, capture, and improve KPIs and a customer-satisfaction index, lifting satisfaction by 2%.
- Created interactive Power BI dashboards to support data-driven decisions at all management levels.
- Implemented ML-based customer segmentation, boosting marketing impact and customer experience by 7.5%.

Bdood.bikes (Shared Bicycle System) Tehran, Iran – Full-time
Head of Data Science & Business Intelligence Team *Apr 2020 – Oct 2020*

- Led the data science and business intelligence team and its analytics roadmap for the bike-sharing operation.
- Implemented dynamic pricing and ML-based usage predictions, increasing revenue and rider experience by 5%.
- Built dashboards with Power BI, Streamlit, and Dash Plotly, boosting engagement by 3%, efficiency by 7%, and station space utilisation by 11%.
- Optimised fleet intelligence, cutting maintenance and relocation costs by 8% and 3% respectively, and raising system performance by 2.7%.
- Led user-behaviour analyses that improved customer retention by 4.5%.

Bdood.bikes (Shared Bicycle System) Tehran, Iran – Full-time
Data Scientist *Jul 2019 – Apr 2020*

- Led predictive-maintenance analysis and battery-health algorithm development, reducing downtime by 3.8% and extending battery life.
- Applied GIS models to identify high-risk riders, enhancing safety by 3% and user experience by 5%.
- Provided marketing and business-analytics insights that contributed to a 6% rise in revenue and engagement.

Additional Professional Development

Generative Modeling Summer School, Eindhoven University of Technology — Eindhoven, The Netherlands
Focus on Deep Generative Modeling — *June 2024*

Machine Learning Summer School, University of Oxford — Oxford, UK
Specialisation in Finance & NLP/Health — *May 2023 – Jul 2023*

Explainable AI (XAI) Summer School, TU Delft — Delft, The Netherlands
Focus on Post-hoc Interpretability in ML — *Aug 2022*

Executive Master in Finance (EMFin), Sharif University of Technology — Tehran, Iran
One-Year Comprehensive Finance Programme — *Nov 2021 – Nov 2022*

Volunteer Activity

Communications Chair of BNAIC/BeNeLearn 2024, Joint International Scientific Conferences on AI and Machine Learning, Utrecht University, 2024.

Skills

Research Interests: Explainable NLP, large language model evaluation, fairness and cultural alignment in LLMs, annotation reliability, human–AI collaboration, open and reproducible research.

Statistical & ML Methods: Bayesian statistics, reinforcement learning and multi-armed bandits (Thompson sampling), large-scale text analysis (LLMs, transformers, topic modelling, LLM-as-judge), data visualisation, geospatial analysis, high-performance computing, privacy-preserving handling of sensitive data.

Programming & Tools: Python (advanced), R, SQL, Bash; Git and GitHub; PyTorch, TensorFlow, Hugging Face, scikit-learn; LLM application stack (RAG, agents, Model Context Protocol, vector databases); Azure ML, MLflow, Docker, CI/CD; Power BI; LaTeX.

Languages: Farsi (native), English (fluent), Dutch (A2, improving).

References

Dr. Robert A. Bagheri, Associate Professor, Department of Methodology & Statistics, Utrecht University — `a.bagheri@uu.nl`

Dr. Anastasia Giachanou, Assistant Professor, Department of Methodology & Statistics, Utrecht University — `a.giachanou@uu.nl`

Prof. dr. Daniel Oberski, Professor of Social Data Science, Department of Methodology & Statistics, Utrecht University; Scientific Director, ODISSEI — `d.l.oberski@uu.nl`